

Customer Segmentation with Clustering

BUSINESS IMPACT AND INSIGHTS

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Introduction and Problem Statement

Understanding different types of customers is important to enhance marketing and pricing strategies, and being customer-centric, leading to customer satisfaction and better customer retention. Effective customer segmentation allows to find customer groups with similar characteristics. The e-commerce organisation is especially interested in grouping customers based on frequency, recency, customer lifetime value (CLV), average cost of the purchases and customer age. Ultimately, this will allow the organisation to use efficient marketing and keep their customers happy, which will maximise revenue.

Methods

The dataset provided includes 951669 rows, which are the products ordered by the customers. It contains 20 columns/features about the customer and the order. We are interested in deriving the following information:

- Frequency: how often a customer purchases
- Recency: time since the last purchase
- Customer lifetime value (CLV): the total value a customer has contributed to the business
- Average unit cost: how much a customer spends on their purchases
- Customer age

Libraries

The required Python were imported: Pandas, Numpy, Matplotlib.pyplot and Seaborn. Specific ML methods were imported from scikit-learn library: RobustScaler, ColumnTransformer, KMeans, silhouette_score, PCA and TSNE. Dendrogram and linkage were imported from Scipy.

EDA and data cleaning

Descriptive statistics, histograms and boxplots were generated to explore the initial variables. The dataset was checked for missing values and duplicates. Duplicated rows were removed.

Feature engineering

New variables Frequency, Recency, CLV, Average_Unit_Cost, and Customer_Age were added to the dataframe. A new dataframe was made with only these five features. The Customer ID was used to aggregate the data by customer. Customer ID was set as the index. To aggregate Frequency, the mean value for sum of purchases was used. For Recency minimum value for days since the last purchase was used. For CLV the sum of the Total revenue was calculated. The average of 'Unit_Cost' was used to calculate the Average Unit Cost. The average age was used to calculate Customer Age. Age, Frequency and Recency were classified as integers. CLV and Average_Unit_Cost were classified as floats with 2 decimal points. After data cleaning and aggregation, there are 68300 rows left, each representing a unique customer. The new features were explored using descriptive statistics, histograms, and boxplots. Because several variables are heavily skewed and contain outliers, RobustScaler was chosen for scaling to reduce the influence of extreme values (Scikit-learn, 2023).

Methods to determine the optimal number of clusters (k)

Three methods were used to determine the optimal number of clusters: the elbow method, silhouette score and dendrogram.

The elbow method

Kmeans was performed with 1-10 clusters and a plot was prepared for the elbow method.

The silhouette method

A sample of the dataframe with 20.000 rows was used to prevent the notebook from crashing. Silhouette scores were calculated for the range between 2-10 clusters.

Hierarchical clustering and dendrogram

Hierarchical clustering was performed and a dendrogram was generated.

Dimensionality reduction and visualisation

Principal Component Analysis and t-SNE were used to compress the features to two components. These were used for 2D visualisation of the output of K-means clustering with k=4. Different values of perplexity were used to generate different t-SNE outputs.

Results

The features 'City', 'Postal_Code', and 'State_Province' show missing values. As these features are not needed for our methods, the missing values do not need adjusting. 21 duplicate rows were found in the dataset and removed. There are no missing values in the dataframe after feature engineering.

EDA of the new features

The histograms in Figure 1 of Frequency, CLV, Recency and Average_Unit_Cost show distributions that are very skewed to the right, with many outliers in the higher values of the boxplot (Figure 2). Recency has an interesting distribution with peaks that seem to repeat every 365 days and then gradually decrease until the next peak. Customer age has a flat distribution, which looks evenly distributed over all age groups, without outliers.

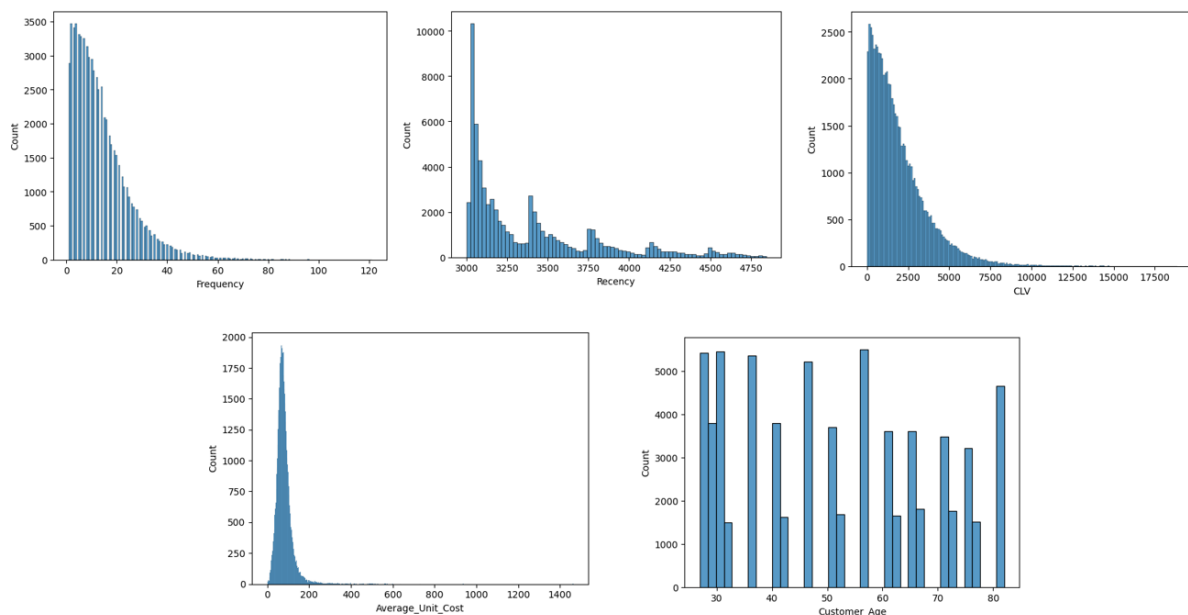


Figure 1. Histograms of the five main features

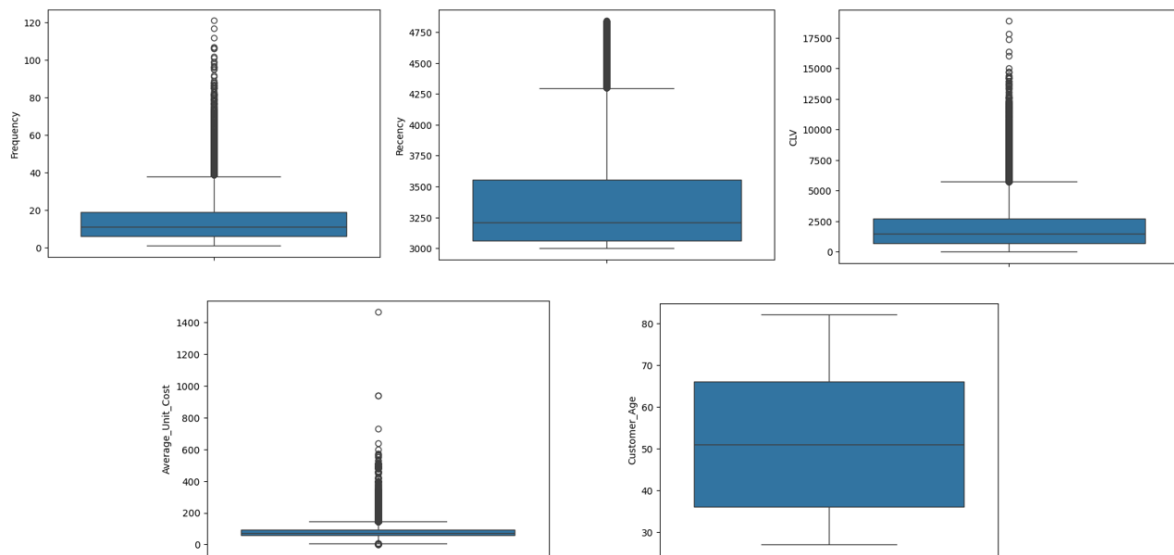


Figure 2. Boxplots of the five main features

The elbow method

Looking at the plot from the Elbow method in **Error! Reference source not found.**, stabilisation of the decline in WCSS can be seen after 6 clusters.

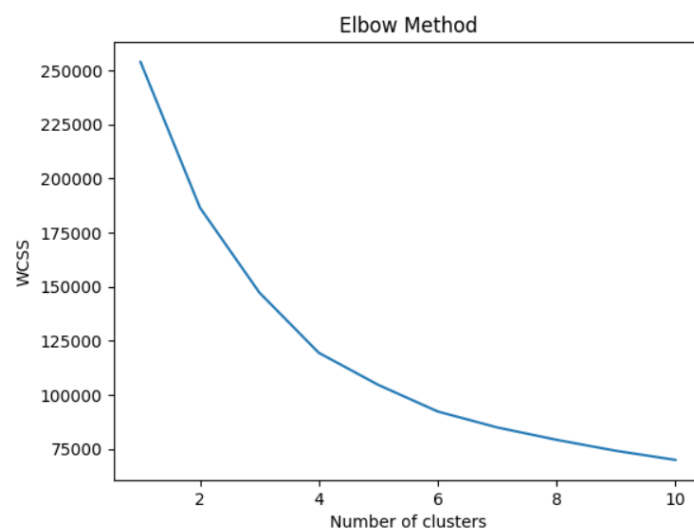


Figure 3. Plot of the Elbow Method

The silhouette method

The number of clusters with the highest silhouette score are 2 clusters (0.27), 3 clusters (0.29) and 4 clusters (0.28). The silhouette score for 6 clusters (0.25).

Silhouette scores for each number of clusters are shown in Table 1.

Table 1. Silhouette scores by number of clusters.

Number of clusters	Silhouette score
2	0.27472564109297226
3	0.2924546558591861
4	0.2828336458035965
5	0.23956951040779803
6	0.2464813904739463
7	0.22606713227412115
8	0.22410705529561045
9	0.22399995303250717
10	0.21201850330752886

The dendrogram

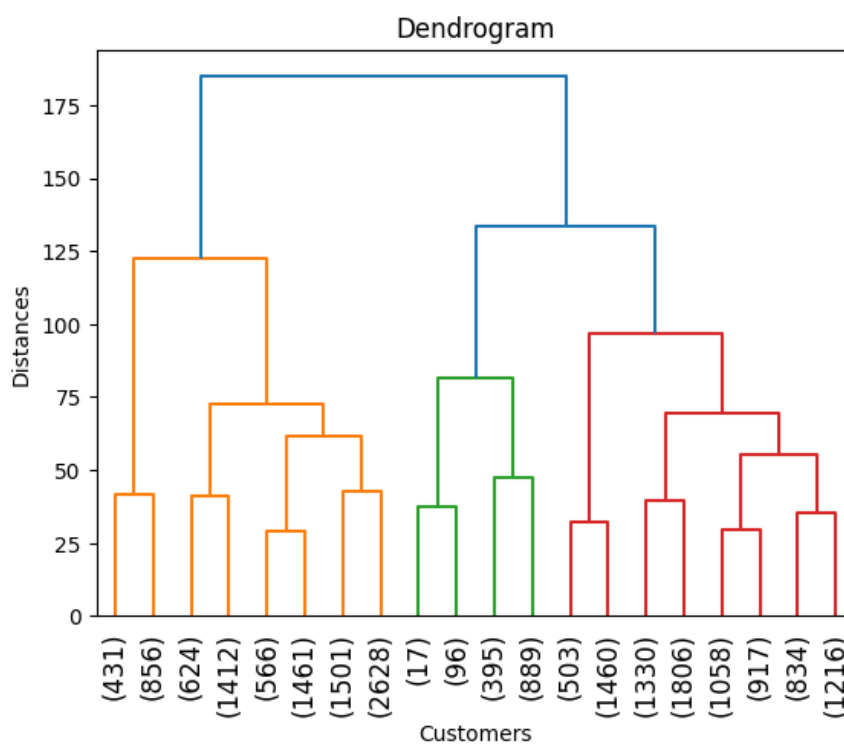


Figure 4. Dendrogram of customers

The cutoff point for 2, 3 or 4 clusters are at very similar distances: respectively at 130, 125 and 120 distances. However, the cutoff point for 6 clusters is around 75, which makes 6 clusters less preferable based on the hierarchical clustering method.

Boxplots of each feature by cluster

From the boxplots in Figure 5 we can observe the following for each cluster: Cluster 3 has a higher average unit cost than the other clusters. Cluster 2 has a higher recency than the other clusters. Cluster 1 has a higher frequency and CLV than the other clusters. Cluster 0 scores low-average compared to the other clusters when it comes to average unit cost, recency, frequency and CLV and has an average age.

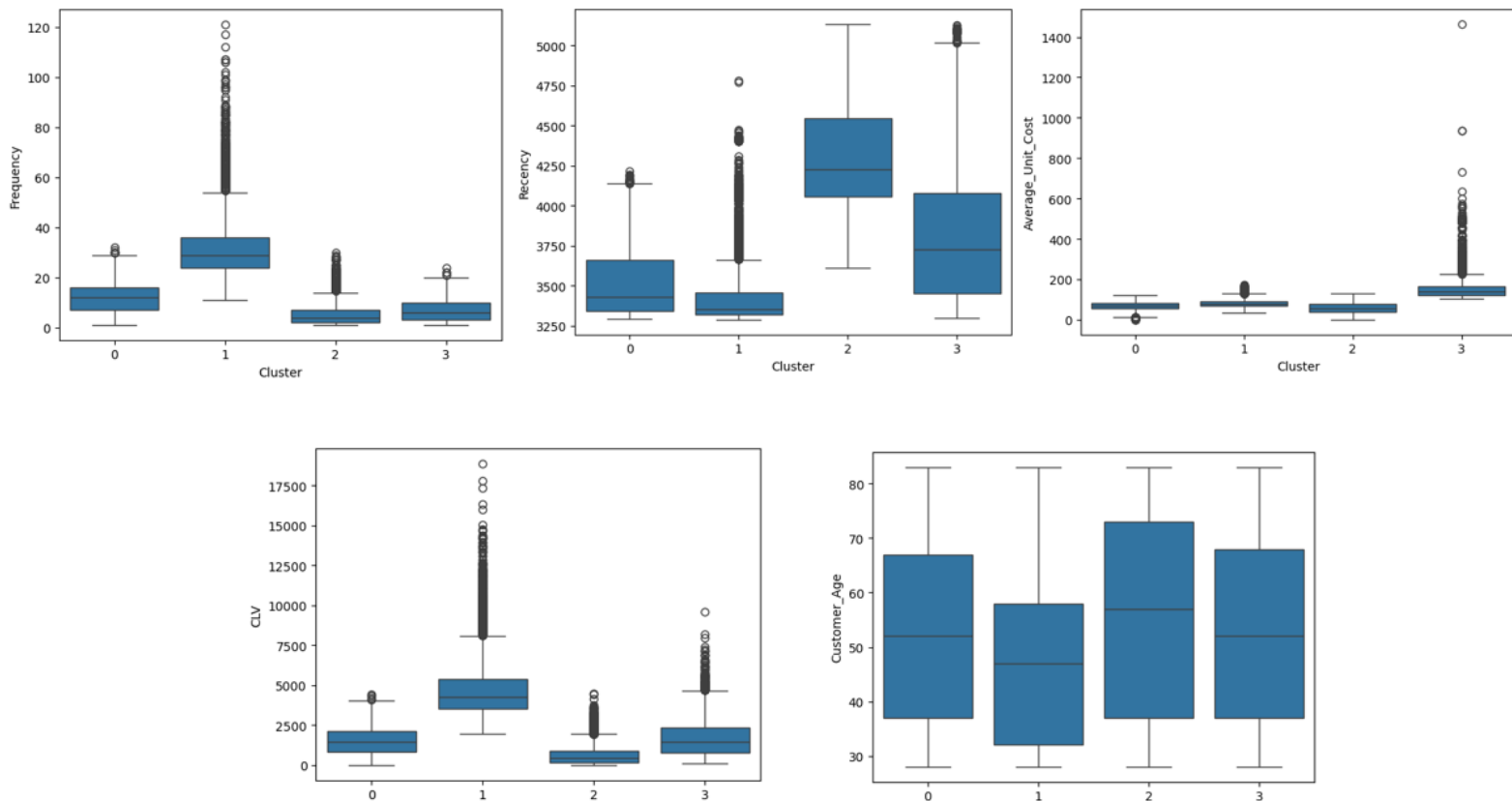


Figure 5. Boxplots of customer clusters

2D Visualisation with PCA and t-SNE

Dimensionality reduction was performed with PCA and t-SNE for visualisation purposes. The visualisation of customer clusters after dimensionality reduction using t-SNE is shown in Figure 7.

2D visualisation of customer clusters after dimensionality reduction with t-SNE (perplexity=10)

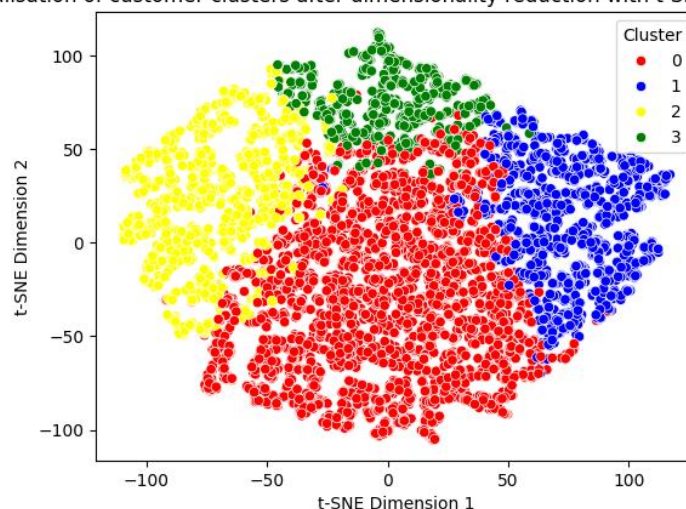


Figure 7. Visualisation of customer clusters after dimensionality reduction with t-SNE

Discussion

Working with unsupervised ML techniques, means that it is unknown what the optimal number of clusters is and there is no absolute right or wrong answer (University of Cambridge, 2023). Using a combination of all three methods, combines the benefits of each method and therefore gives a better overview to determine the optimal number of clusters.

In this case the number of 4 clusters was selected. This is in the middle of the Elbow, slightly higher than 6 clusters. However, the silhouette score and dendrogram show that 2, 3, or 4 clusters are preferred over 5 or 6 clusters.

Cluster 3 can be seen as customers who spend more per order, who have a higher average unit cost. Cluster 2 can be seen as recent customers, who score clearly higher on Recency. Cluster 1 customers have a higher frequency and CLV. This is an important group as they are the frequent buyers who have spent more in the long-term. The first group of customers (Cluster 0) do not score high on frequency, recency, average unit cost or CLV. These seem to be the rest of the customers, who do not fall under another cluster.

Conclusion

The Elbow method, silhouette score and hierarchical clustering using a dendrogram are effective ways to determine the number of customer clusters. Using the three methods combined, provides an excellent way to determine the optimal number of clusters. In this case, four clusters of customers were defined with different characteristics, which can enhance targeted marketing strategies and ultimately increase revenue.

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